

Best Subset for Predicting Private Undergraduate Institution In-State Tuition

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Read in Data

Read csv from this link into a dataframe `df`.

Code book available [here](#).

Each row is unique by column UNITID.

1. Data Cleaning

For this analysis, we will focus on only Private school.

```
##           CONTROL  min  median   max      sd  n missing
## 1 Private for-profit 5160 16449.0 51400 7555.020 80      14
## 2 Private not-for-profit    0 38400.0 85000 16047.532 778      6
## 3           Public   910  7019.5 21989  3804.278 758      4
```

```
## Number of Private schools = 858
```

```
## Number of columns = 122
```

1st Cleaning

Next cleaning will does the following.

- filter the schools down to only Private for-profit and Private not-for-profit
- create dummy variables to distinguish between for-profit and not-for-profit
- turn any categorical variables into dummy variables (if there are n levels, there will be n-1 extra variables)
- remove the original categorical variables because we will use the created dummies for analysis from now on.

```
## Number of columns after 1st cleaning = 135
```

2nd Cleaning

Check NAs, we don't want to use variables that have a lot of missing rows (over 5%). Variables like Men, Women we will only keep Men. We will also remove other tuition data like out-of-state tuition, published tuition, and other uninterested variables.

```
## Number of columns after 2nd cleaning = 67
```

3rd Cleaning

We will now do a third time cleaning:

- Remove any non-numerical variables (to remove extra categorical variables we're not interested in. If the number of columns decrease, should check what extra columns got removed). In this case, we removed ATHASSOC (Member of National Athletic Association) because we have one for NCAA already, removed CARNEGIEIC (Institutional Classification) because we have program mix and this variable is over-complicated.
- Remove any row with NAs

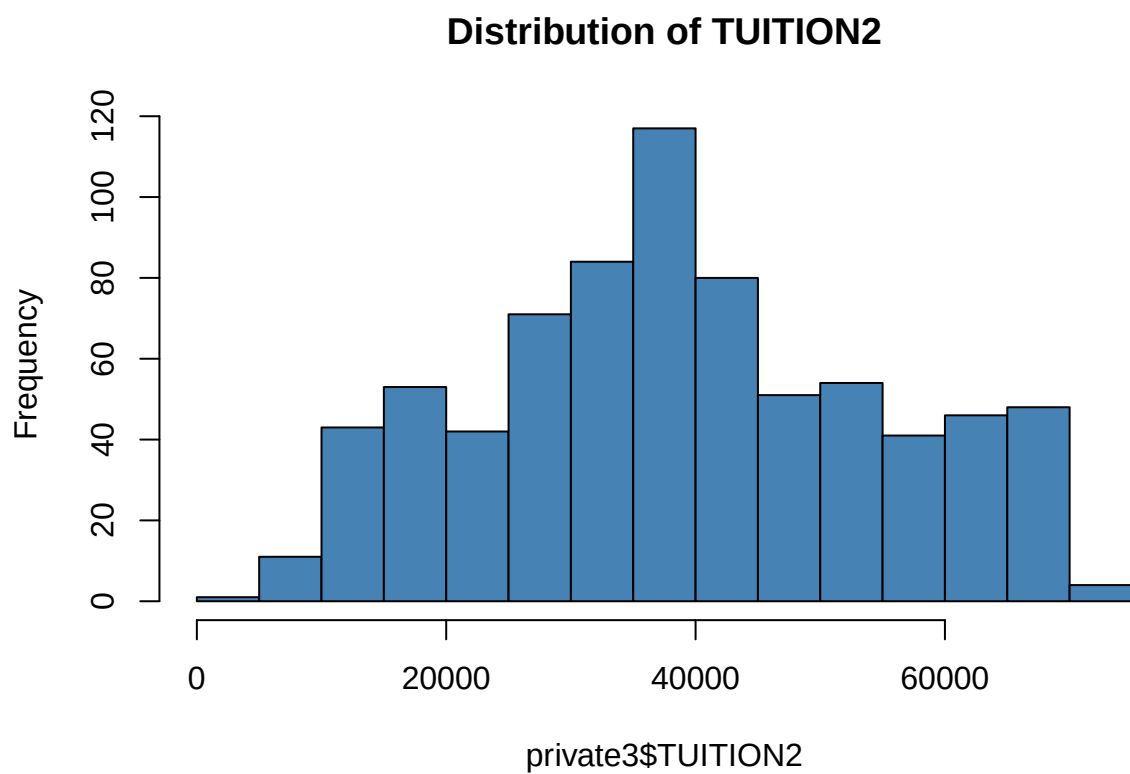
```
## Number of rows after 3rd cleaning = 746
```

```
## Number of columns after 3rd cleaning = 65
```

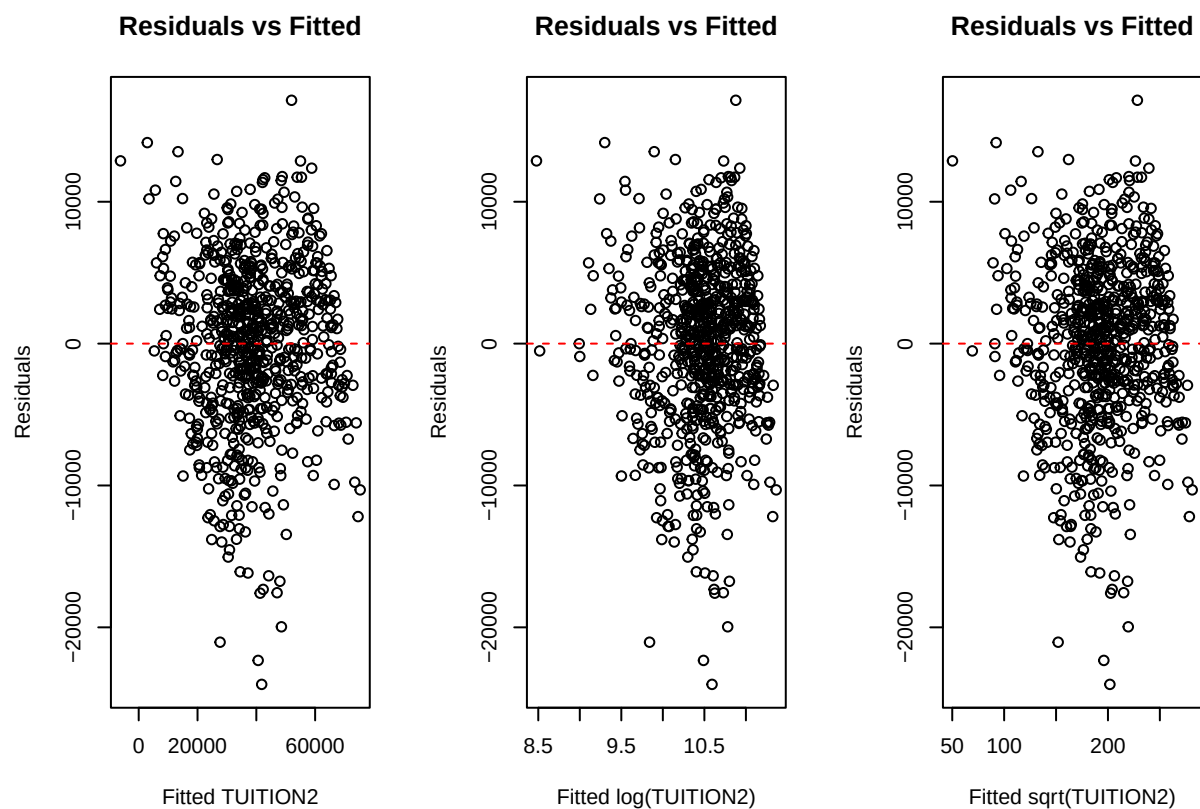
2. Data Transformation

Response Variable: TUITION2

Tuition price is almost always right-skewed and usually need log-transformation but we will check just in case. Here's the distribution of the original TUITION2.



The histogram looks fairly normal and bell-shaped. Just as a sanity check, we will check if doing log-transformation and square-root-transformation have good impact on the distribution of the data or not.

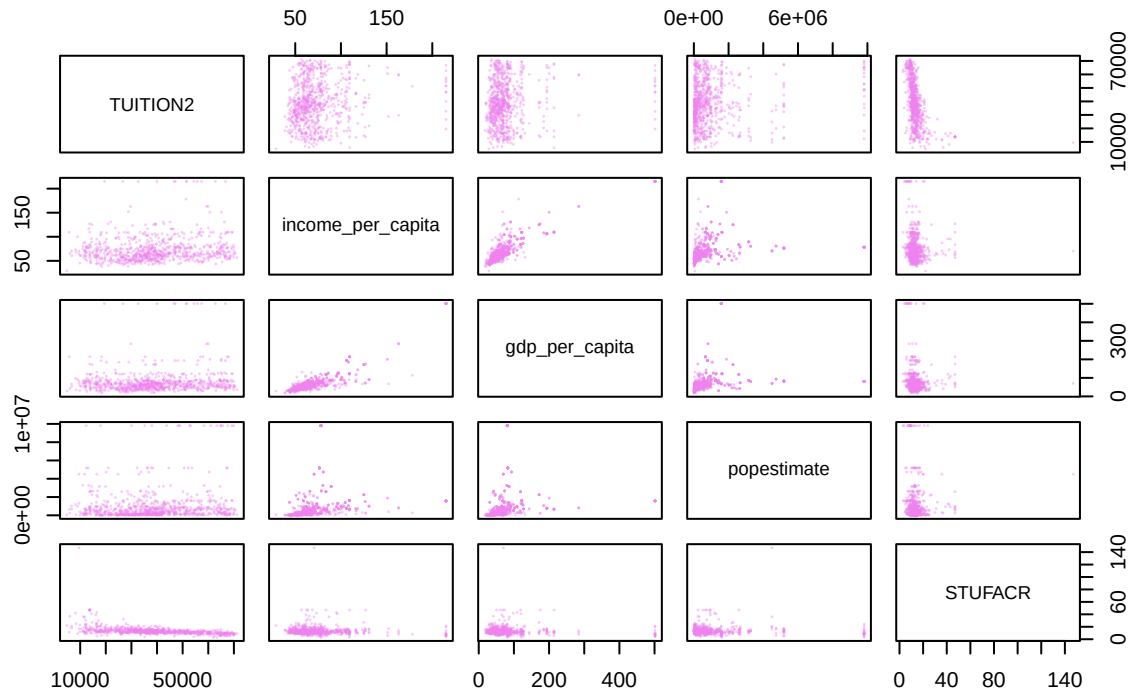


The residual plot with the **original TUITION2** looks fairly random with mild heteroskedasticity (variance increases with fitted values). The other two plots are not much different. We want to keep our model simple, thus, we will continue to use the original TUITION2 as our response variable.

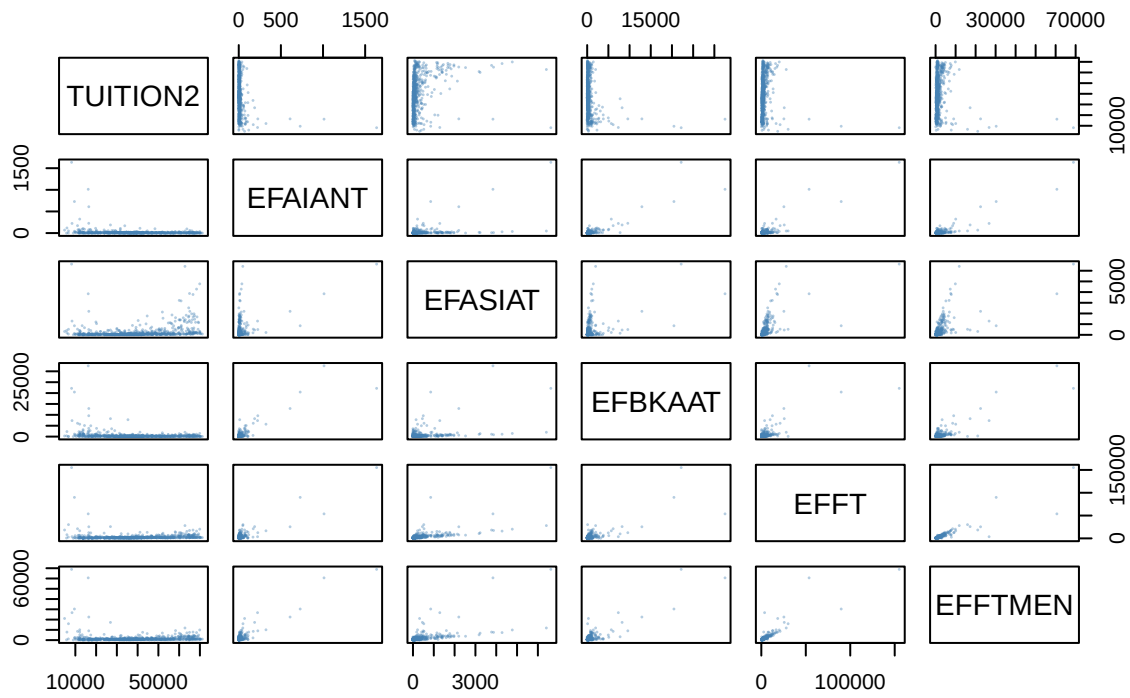
Numeric variables vs TUITION2

We will now check numerical variables in pair plots with TUITION2 to see if we need to perform transformations.

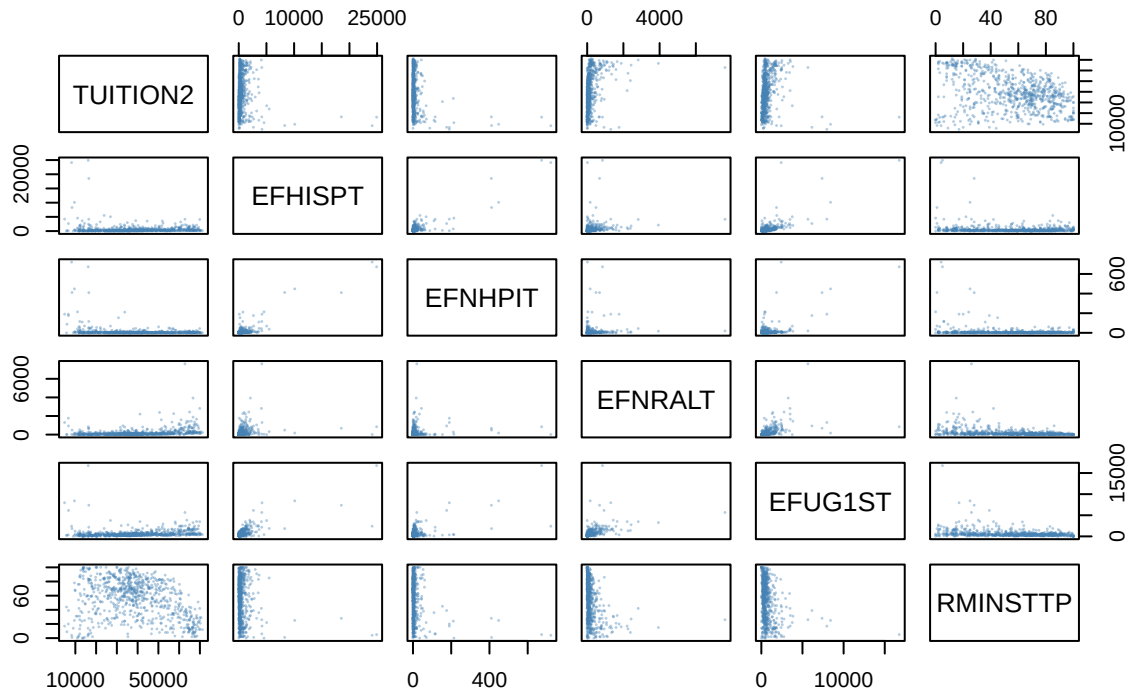
Matrix Plot: Institution Profile & Resources vs In-State Tuition



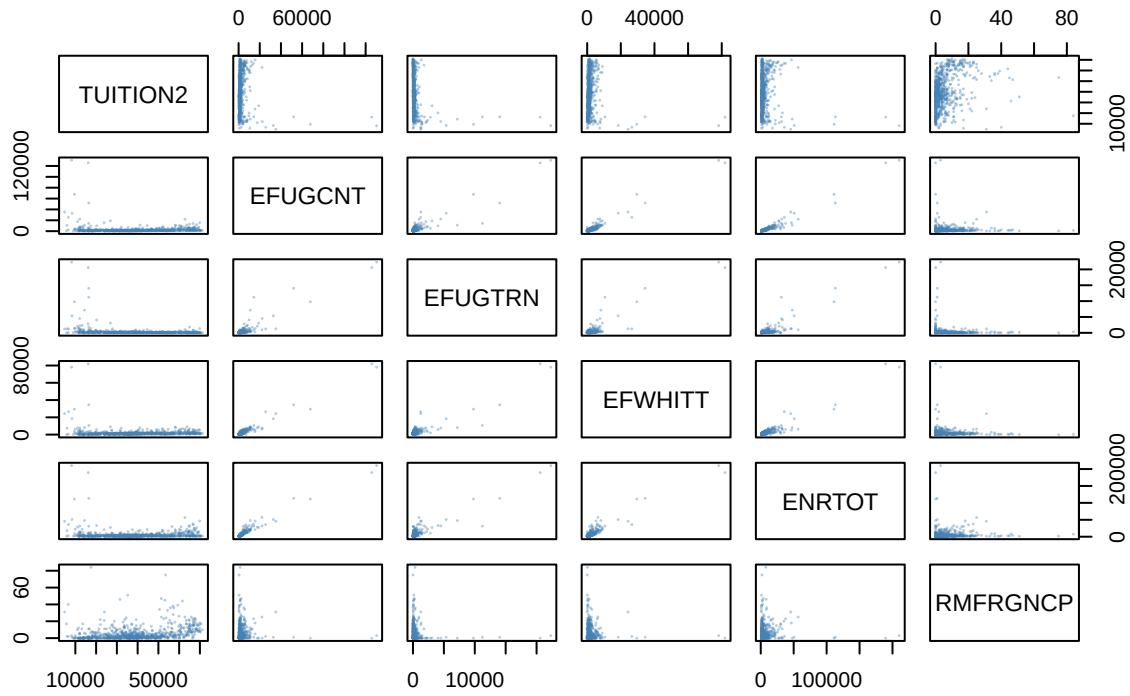
Matrix Plot: Enrollment Variables 1 vs In-State Tuition



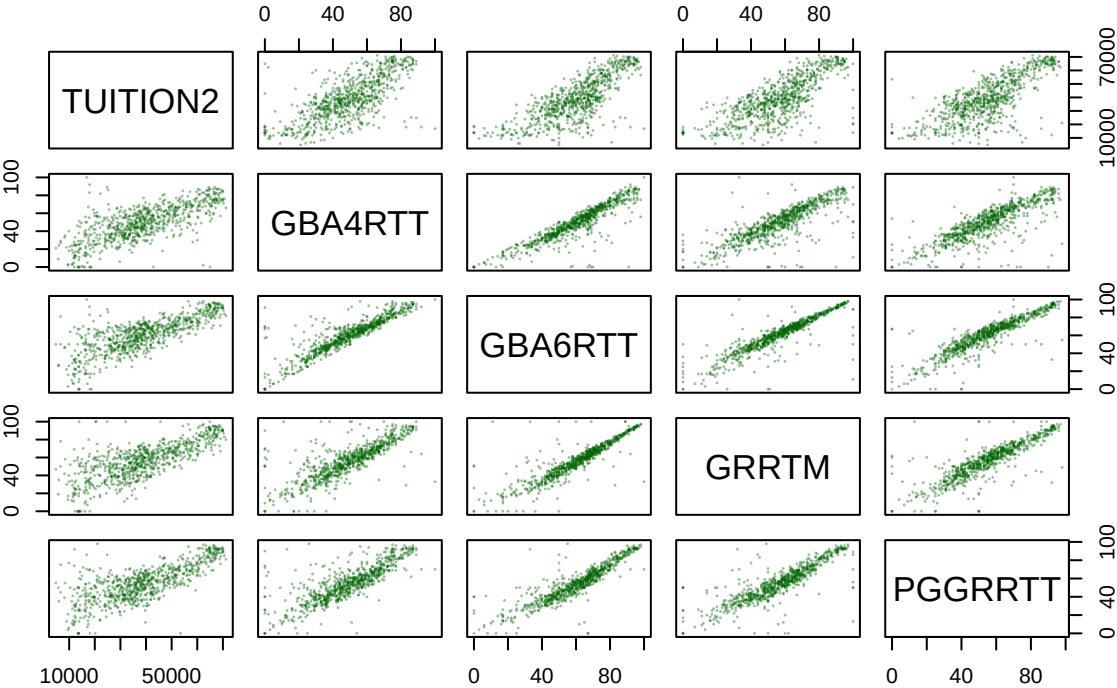
Matrix Plot: Enrollment Variables 2 vs In-State Tuition

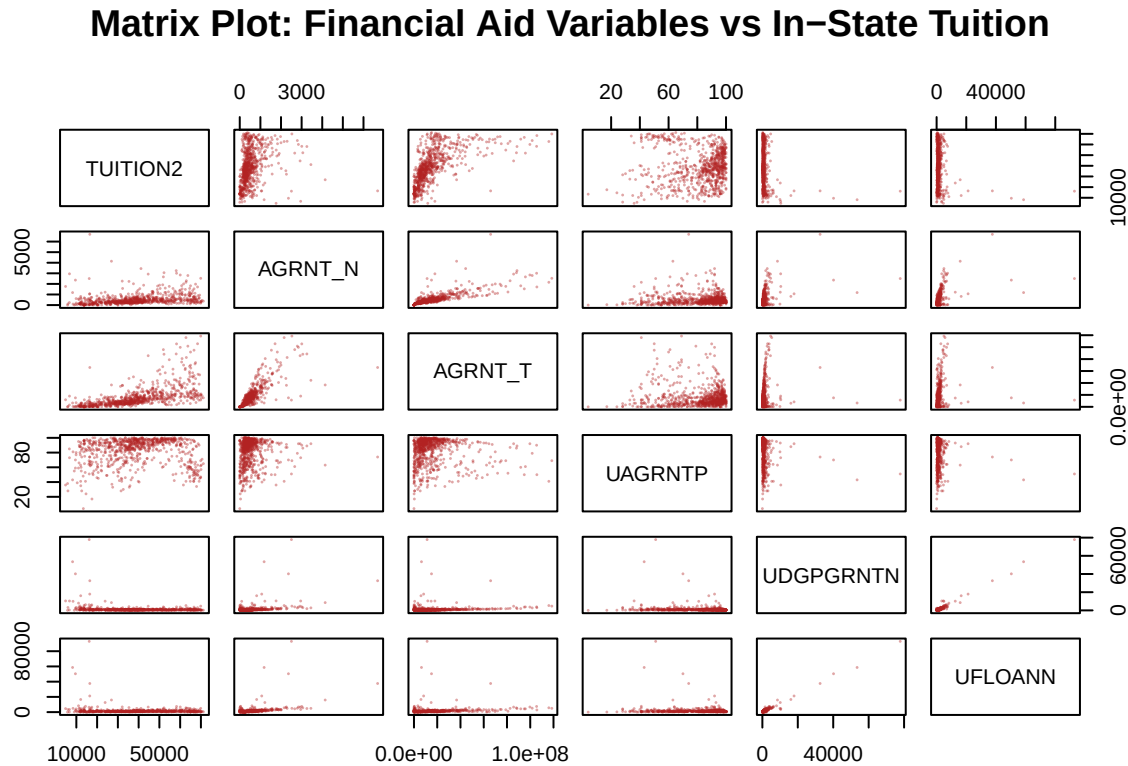


Matrix Plot: Enrollment Variables 3 vs In-State Tuition



Matrix Plot: Admissions & Outcomes vs In-State Tuition





Most of our numerical variables have extreme right skewness. Thus we will:

- Log-transform $\log(x + 1)$ on counts or total dollar amounts (Enrollment & demographic counts, grants, aid and economic resources) since they are heavily right-skewed
- Square-root transformation $\sqrt{x}$ on ratios like STUFACR since it has moderate skewness
- No transformation on variables like graduation rates (GBA4RTT, GBA6RTT, GRRTM, PGGRRTT) and certain percentages (UAGRNT, RMINSTP) since they appear relatively normally distributed or well-spread

The transformed numerical variables pair plots look much better than that of the original numerical variables in the following pages. The plots will not be included in the knitted version but the code is available in the Rmd.

3. Best Subset Selection

Group Variables into Categories

We have too many predictors to run one model. So we split into groups as following (Note that this grouping is strictly for best subset and grouping in the R Shiny App should follow the code book).

- Group 1: Enrollment & Demographics
- Group 2: Outcomes & Institutional Characteristics

- Group 3: Financial Aid & Grant
- Group 4: Institutional / Structural Indicators
- Group 5: Geography
- Group 6: Program Mix

```
## Manual groupings count matches nubmer of columns in private4: TRUE
```

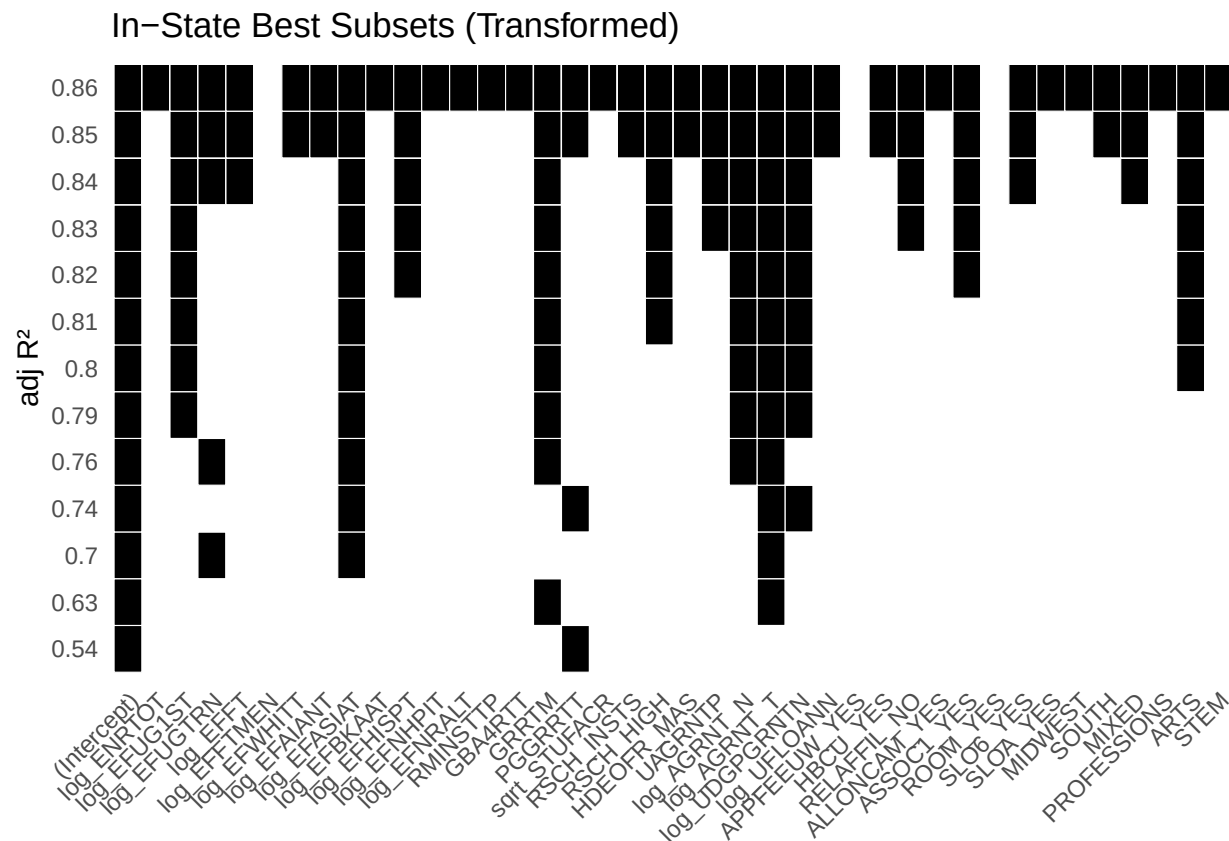
Next, we run best subset on each group and find the best variables from each group:

We also run into a potential problem that is a variable could be not important in its grouping best subset but could have value when paired with others. Right now, our analysis does *not* take care of this issue and does *not* consider interactions. The most important variables are in the pool (lower bic, better model):

```
## [1] "log_ENRTOT"      "log_EFUG1ST"    "log_EFUGTRN"    "log_EFFT"
## [5] "log_EFFTMEN"     "log_EFWHITT"    "log_EFAIANT"    "log_EFASIAT"
## [9] "log_EFBKAAT"     "log_EFHISPT"    "log_EFNHPIT"    "log_EFNRLT"
## [13] "RMINSTTP"        "GBA4RTT"        "GRRTM"          "PGGRRTT"
## [17] "sqrt_STUFACR"    "RSCH_INSTS"     "RSCH_HIGH"      "HDEOFR_MAS"
## [21] "UAGRNTNTP"       "log_AGRNT_N"    "log_AGRNT_T"    "log_UDGPGRNTN"
## [25] "log_UFLOANN"     "APPFEEUW_YES"   "HBCU_YES"       "RELAFFIL_NO"
## [29] "ALLONCAM_YES"    "ASSOC1_YES"     "ROOM_YES"       "SLO6_YES"
## [33] "SLOA_YES"        "MIDWEST"        "SOUTH"          "MIXED"
## [37] "PROFESSIONS"     "ARTS"           "STEM"
```

Final Best Subset

Now, we run another best subset with the variables we deem important from the above analysis.



Linear Model

We now create a linear model with the selected variables from the best subset. We will include all n-1 levels of a categorical variable we transformed into dummies.

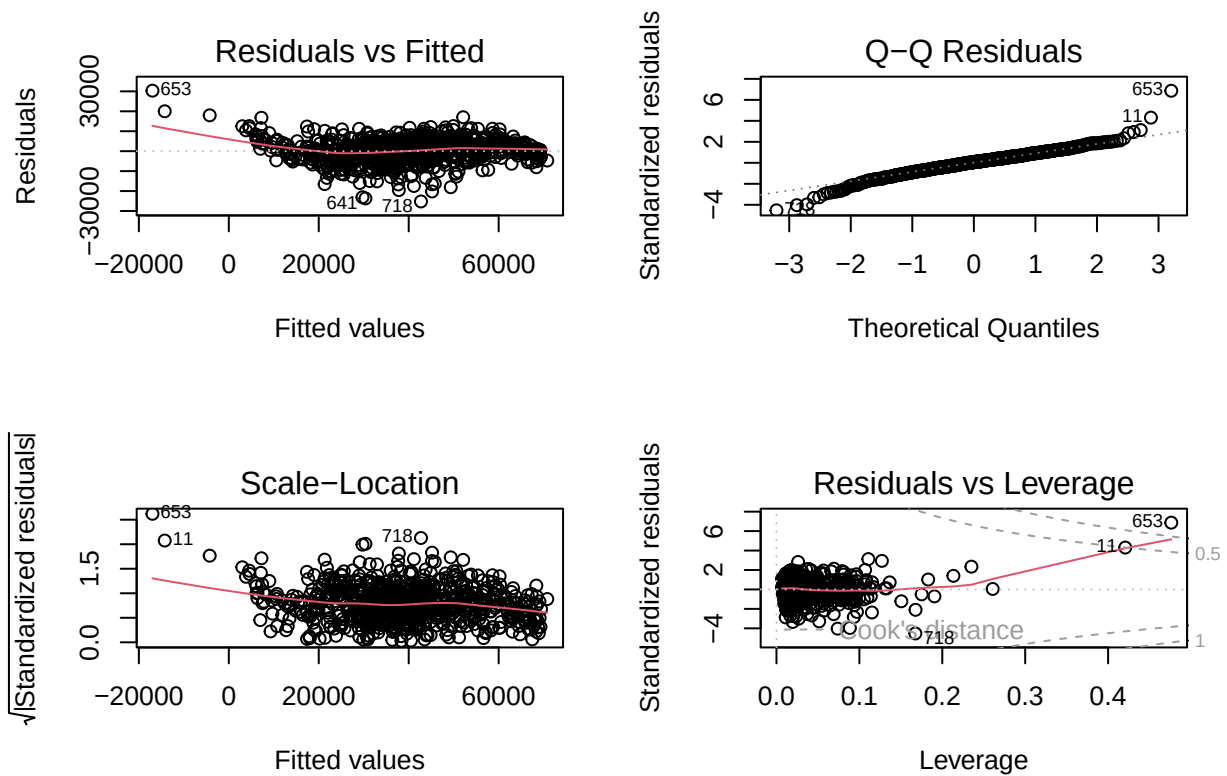
```
##
## Call:
## lm(formula = TUITION2 ~ log_ENRTOT + log_EFUG1ST + log_EFUGCNT +
##     log_EFASIAT + log_EFHISPT + log_EFWHITT + log_EFNHPIT + GBA6RTT +
##     GRRTM + sqrt_STUFACR + RSCH_HIGH + RSCH_INSTS + log_AGRNT_N +
##     log_AGRNT_T + log_UDGPGRNTN + log_UFLOANN + HBCU_YES + RELAFFIL_NO +
##     ASSOC1_YES + SLO6_YES + MIXED + HEALTH + PROFESSIONS + ARTS +
##     STEM + FORPROFIT - UNITID, data = private4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25208.0  -3423.6   386.1   3732.5  30338.7
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -37298.07    5630.38  -6.624 6.83e-11 ***
## log_ENRTOT    -3247.45     817.59  -3.972 7.85e-05 ***
## log_EFUG1ST    3644.06     859.63   4.239 2.54e-05 ***
## log_EFUGCNT    7595.74    1223.82   6.207 9.15e-10 ***
```

```

## log_EFASIAT      1743.79      334.15      5.219 2.36e-07 ***
## log_EFHISPT      1039.62      315.92      3.291 0.001048 **
## log_EFWHITT      -2298.54      464.42     -4.949 9.28e-07 ***
## log_EFNHPIT      -879.03      277.30     -3.170 0.001589 **
## GBA6RTT           23.23       30.10      0.772 0.440488
## GRRTM             88.31       26.44      3.341 0.000879 ***
## sqrt_STUFACR      250.31      484.46      0.517 0.605531
## RSCH_HIGH         7352.76     1085.87      6.771 2.65e-11 ***
## RSCH_INSTS        2352.59      968.20      2.430 0.015348 *
## log_AGRNT_N       -9641.91      890.55    -10.827 < 2e-16 ***
## log_AGRNT_T        7690.50      469.47     16.381 < 2e-16 ***
## log_UDGPGRNTN     -8797.93      786.74    -11.183 < 2e-16 ***
## log_UFLOANN        713.57      385.32      1.852 0.064451 .
## HBCU_YES          -15215.98     2708.54     -5.618 2.77e-08 ***
## RELAFFIL_NO       1496.82      533.85      2.804 0.005186 **
## ASSOC1_YES        2482.03      620.39      4.001 6.97e-05 ***
## SLO6_YES          3660.46      982.71      3.725 0.000211 ***
## MIXED             2389.81     1254.31      1.905 0.057142 .
## HEALTH            -100.79     1413.83     -0.071 0.943187
## PROFESSIONS       -208.33     1159.72     -0.180 0.857485
## ARTS              5981.81     1300.41      4.600 4.99e-06 ***
## STEM              -842.51     2011.53     -0.419 0.675456
## FORPROFIT         1474.33     1363.18      1.082 0.279821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6105 on 719 degrees of freedom
## Multiple R-squared:  0.856, Adjusted R-squared:  0.8508
## F-statistic: 164.3 on 26 and 719 DF, p-value: < 2.2e-16

```

We also generate plots to validate the model.



The model suggests that private tuition is a function of “Prestige and Policy” rather than “Place.” The high significance of variables like RSCH_HIGH (Research status), SLO6_YES (Study Abroad), and the complex interplay of grant totals (log_AGRNT_T) vs. recipient counts (log_AGRNT_N) points to a high-cost/high-aid model.

Essentially, private schools set tuition based on the amenities and institutional prestige they provide (like research and study abroad) and their ability to “price discriminate” through financial aid, rather than the local cost of living in their specific county.

Next step: justifying outliers, validating models, run this for out-of-state tuition, public schools, etc.